

DUNMAN SECONDARY SCHOOL

CANDIDATE
NAME

CLASS

INDEX
NUMBER

PRELIMINARY EXAMINATION 2023 SECONDARY 4 NORMAL ACADEMIC

MATHEMATICS SYLLABUS A

Paper 1

4045/01

27 July 2023

2 hours

Marking Scheme

Question	Answer	Marks
1	$\frac{20 \times 0.01}{\sqrt{100}}$	2
	$= \frac{0.2}{10}$	
	$= 0.02$	

Question	Answer	Marks
2	$x \geq -8$	2
	2	

Question	Answer	Marks
3	$(2^2)^{\frac{5}{2}} \div (2^3)^3 = 2^n$	2
	$2^5 \div 2^9 = 2^n$	
	$n = -4$	

Question	Answer	Marks
4	$\frac{q-1}{p-2} = \frac{4}{5}$	2
	$= \frac{12}{15}$	
	$p = 17, q = 13$	
	$\therefore (17, 13)$	
Alternative Method	$y = \frac{4}{5}x - \frac{3}{5}$	
	Since p and q must be positive integers more than 10 but not more than 20,	
	$p = 17, q = 13$	
	$\therefore (17, 13)$	

Question	Answer	Marks
5a	$\frac{1}{3}$	1
5b	$\frac{2}{3}$	1

Question	Answer	Marks
6a	27	1
6b	45.5	1

Question	Answer	Marks
7a	$(24)(50) = 1200 \text{ cm}$	1
	$= 12 \text{ m}$	
7b	$1 \text{ cm}^2 : 0.25 \text{ m}^2$	2
	$10.2 \text{ m}^2 \text{ represents } 10.2 \div 0.25$	
	$= 40.8 \text{ cm}^2$	

Question	Answer	Marks
8	$40.5^\circ (+ 360n) \text{ or } 139.5^\circ (+ 360n) (1 \text{ d.p.})$	2

Question	Answer	Marks
9a	$x = \frac{-6+1}{2}$	2
	$x = -2.5$	
9b	$y = -(-2.5-1)(-2.5+6) = 12.25$	1
	$\therefore (-2.5, 12.25)$	

Question	Answer	Marks
10a	Time = $\frac{28 \text{ km}}{48 \text{ km/h}}$	2
	$= \frac{7}{12} \text{ h}$	
	$= 35 \text{ minutes}$	
10b	$\frac{48 \text{ km}}{1 \text{ h}} = \frac{48000 \text{ m}}{3600 \text{ s}}$	1
	$= 13\frac{1}{3} \text{ or } 13.3 \text{ m/s (3 s.f.)}$	

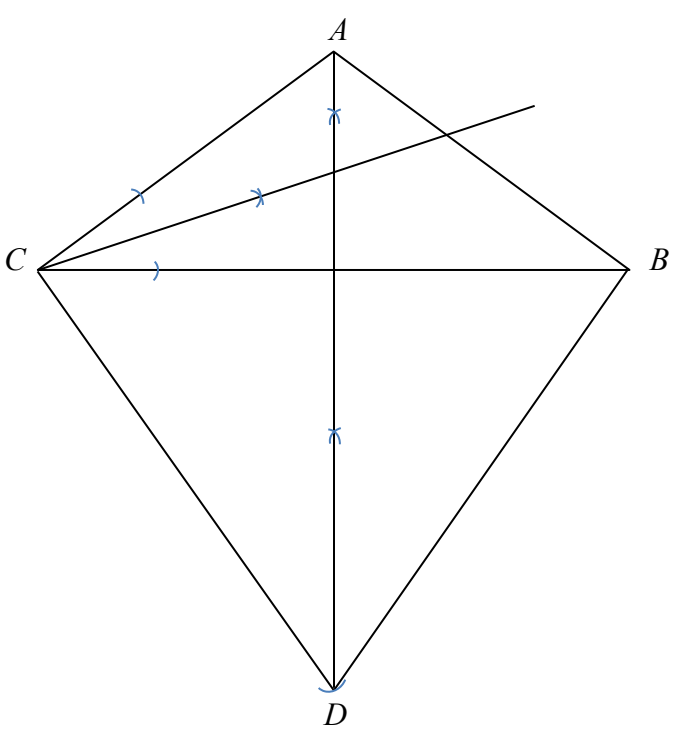
Question	Answer	Marks
11a	85% rep. \$1275	2
	100% rep. $\frac{100}{85} \times 1275$	
	$= \$1500$	
11b	$\frac{98-70}{70} \times 100\%$	2
	$= 40\%$	

Question	Answer	Marks
12a	$a = \underline{\mathbf{238}}$	2
	because <u>the sum of angles at a point is 360°.</u>	
12b	$b = 180^\circ - [(180^\circ - 132^\circ) + (180^\circ - 122^\circ)]$	2
	$= 74^\circ$	

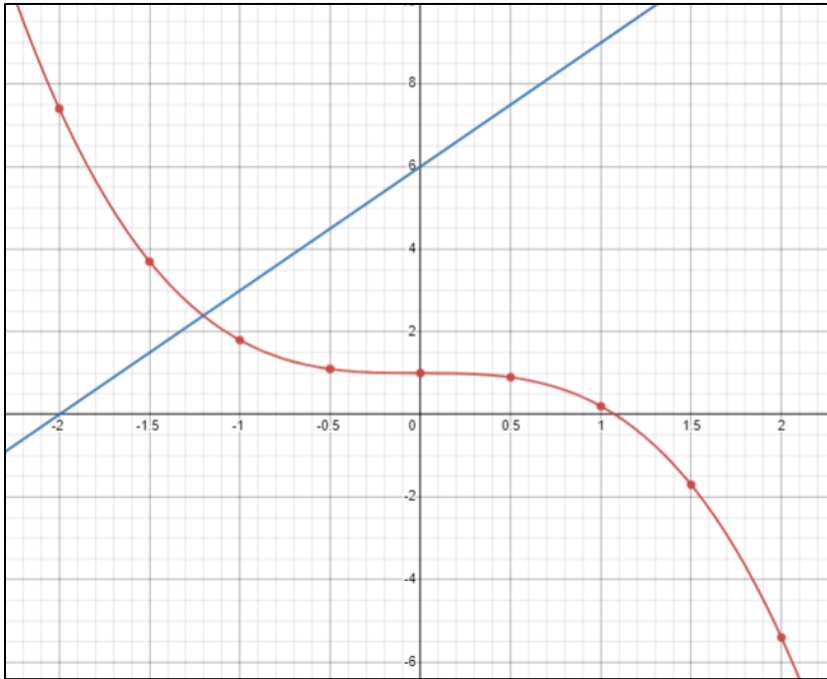
Question	Answer	Marks
13a	$h = 6$	2
	$k = -11$	
13b	$(x+6)^2 - 11 = 0$	2
	$(x+6)^2 = 11$	
	$x = -6 \pm \sqrt{11}$	
	$x = -2.68$ or -9.32 (2 d.p.)	

Question	Answer	Marks
14a	$p^2 + (p-14)^2 = (p+2)^2$	2
	$p^2 + p^2 - 28p + 196 = p^2 + 4p + 4$	
	$p^2 - 32p + 192 = 0$ (shown)	
14b	$(p-24)(p-8) = 0$	3
	$p = 8$ (rej.) or 24	
	$AB = 24$	

Question	Answer	Marks					
15a		1					
	Pattern number		1	2	3	4	5
	Grey tiles		1	2	3	4	5
	White tiles		0	2	4	6	8
	Total		1	4	7	10	13
15bi	$2n - 2$	1					
15bii	$3n - 2$	1					
15c	$3n - 2 = 133$	1					
	$n = 45$						

Question	Answer	Marks
16a		1
16b		1
16c		1

Question	Answer	Marks
17	$\frac{360^\circ}{8}$	2
	$= 45^\circ$ By alt. angles of parallel lines, angle $CPD = 45^\circ$	
	$\text{Interior angle} = \frac{(8-2) \times 180^\circ}{8}$	
	$= 135^\circ$	
	$\text{Angle } CPD = 180^\circ - 135^\circ$	
	$= 45^\circ$	

Question	Answer	Marks												
18a	<table><tr><td>x</td><td>0</td><td>0.5</td><td>1</td><td>1.5</td><td>2</td></tr><tr><td>y</td><td>1</td><td>0.9</td><td>0.2</td><td>-1.7</td><td>-5.4</td></tr></table> 	x	0	0.5	1	1.5	2	y	1	0.9	0.2	-1.7	-5.4	4
x	0	0.5	1	1.5	2									
y	1	0.9	0.2	-1.7	-5.4									
18b		2												
18c	$(-1.2, 2.4)$	1												

Question	Answer	Marks
19a	$AB = \sqrt{(2 - (-1))^2 + (-4 - (-8))^2}$	2
	$= \sqrt{3^2 + 4^2}$	
	$= 5 \text{ units}$	
19b	Gradient $= \frac{3}{4}$	2
	$y = \frac{3}{4}x + 5$	
Alternative method	$y = \frac{3}{4}x + 5$	
19c	$(-3, -1)$	1
19d	$(5)(3)$	1
	$= 15 \text{ units}^2$	

Question	Answer	Marks
20a	Although white bags take up a larger proportion for Ken as compared to Wendy's, the total number of bags sold by each of them may be different.	1
20b	15% rep. 42 bags	1
	100% rep. $\frac{100}{15} \times 42$	
	$= 280 \text{ bags}$	
20c	It is easier to compare proportion/ relative size of each colour sold , especially in comparison to one quarter or half.	1

Question	Answer	Marks
21a	Since $AC \parallel DE$ and $AC = DE$, $ACDE$ is a parallelogram.	2
	$\therefore AE \parallel CD \therefore ABDE$ is a trapezium as it has a pair of parallel lines.	
21b	$\frac{1}{2}(18 + 34)(15)(k) = 19500$	3
	$k = 50$	